

Full Marking Scheme

Autumn 2024 Examination · Paper 1

Study-Uni · Full Marking Scheme

Institution	University College Cork (National University of Ireland, Cork)
External Examiner	Prof. Mac an Bhaird
Head of Department	Prof. Doran
Internal Examiner	Dr. Turner
Structure	Section A (Long Questions): answer 2 of 3 questions, each carrying 100% of that question's marks
Duration	1.5 hours
Materials Permitted	Non-programmable calculators only

Coverage: this scheme answers all three Section A questions (A1, A2, A3) in full, even though the exam instructions only require you to attempt two of them. Work through all three during revision, then practise choosing your best two under timed conditions.

Source basis: the statistical tables (Normal, t, F, Durbin-Watson) and the exam question text below are reproduced verbatim from the official Autumn 2024 EC4215 paper. The testing procedures used in the answers (t-test, F-test for overall significance, the four specification criteria for variable inclusion, the omitted-variable-bias result, and the Breusch-Pagan / White / Goldfeld-Quandt heteroscedasticity tests) are drawn directly from the EC4215 lecture material you uploaded. All numeric results below (t-ratios, the F statistic, and its critical value) were independently computed and verified against those formulae — see the worked tables in each answer.

Question A1

Interpreting a multiple regression & testing overall significance

(a) Suppose that a researcher is examining the earnings of a rock band over time and her data produces the following econometric regression results (standard errors in brackets):

$$E_t = 50.3 + 0.65D_t + 5.2T_t + 65.1A_t \quad (0.17) \quad (1.8) \quad (18.3)$$

$$\bar{R}^2 = 0.78 \quad n = 18$$

where E_t = earnings (€000), D_t = downloads (000s), T_t = tickets sold (000s), $A_t = 1$ if a new album was released in year t , 0 otherwise.

- | | |
|--|------------|
| (i) Carefully interpret the above findings. | 35% |
| (ii) Does the above equation fit your expectations? Why or why not? | 15% |
| (iii) If you could add one more variable into the above regression, what would it be and why? | 10% |

Intercept (50.3): when $D_t = T_t = 0$ and no album is released ($A_t = 0$), the model predicts baseline earnings of €50,300. In practice this captures earnings from sources not in the model (royalty back-catalogue, sponsorship, etc.) rather than a literal "zero activity" scenario, so it should be interpreted cautiously rather than taken as a meaningful standalone figure.

Downloads, D_t ($\hat{b} = 0.65$): holding T_t and A_t constant, each additional 1,000 song downloads is associated with a €650 increase in the band's earnings that year. The coefficient's t-ratio is $0.65/0.17 = 3.82$, which exceeds the critical t-value for 14 degrees of freedom at the 5% level (2.145), so downloads have a statistically significant, positive effect on earnings.

Tickets sold, T_t ($\hat{b} = 5.2$): holding D_t and A_t constant, each additional 1,000 tickets sold is associated with a €5,200 increase in earnings — roughly eight times the per-unit effect of downloads. This is economically sensible: a single concert ticket carries a much higher price point (and generates ancillary revenue such as venue splits and merchandise) than a single download. The t-ratio is $5.2/1.8 = 2.89$, significant at 5%.

Album release, A_t ($\hat{b} = 65.1$): holding downloads and ticket sales constant, releasing a new album in a given year is associated with €65,100 higher earnings than a year with no release. The t-ratio is $65.1/18.3 = 3.56$, significant at 5%. This effect is large relative to the other two, consistent with a new album driving a broad uplift (press, streaming, tour demand) that isn't fully captured by that year's download and ticket figures alone.

Goodness of fit: $\bar{R}^2 = 0.78$ means that, after adjusting for the number of regressors and the small sample size ($n = 18$), the three explanatory variables jointly explain 78% of the year-to-year variation in the band's earnings — a strong fit for cross-time earnings data with only three predictors.

Working — t-ratios for Part (a)(i)

Variable	Coefficient	Std. Error	t = $\hat{b}/se$	Significant at 5%? (df = 14, $t_{crit} = 2.145$)
D_t (downloads)	0.65	0.17	3.824	Yes
T_t (tickets)	5.2	1.8	2.889	Yes
A_t (album dummy)	65.1	18.3	3.557	Yes

SOURCING NOTE

The regression-interpretation approach (coefficient sign, magnitude "holding other variables constant", and comparing the t-ratio to t_{crit} from the paper's own Table A3) follows standard OLS interpretation covered in the EC4215 lecture material; $df = n - k - 1 = 18 - 3 - 1 = 14$.

FULL MODEL ANSWER — PART (A)(II)

Broadly, yes — the equation matches sensible economic expectations. All three coefficients carry the sign an economist would predict: more downloads, more ticket sales, and a new album release should all raise a band's earnings, and each estimated effect is positive and statistically significant. The relative magnitudes are also plausible: ticket revenue per unit and the one-off album effect both dwarf the per-download effect, which matches how the music industry actually monetises listeners — live performance and album cycles generate far more revenue per event than an individual digital download.

That said, two things are worth flagging as not fully matching expectations. First, the intercept of €50,300 is large for a "no activity" baseline, hinting that some relevant revenue source (e.g. royalties, sponsorship, merchandise) is being absorbed into the constant rather than modelled explicitly. Second, with only $n = 18$ observations the standard errors are relatively imprecise, so while the point estimates have sensible signs and magnitudes, the confidence intervals around them are correspondingly wide.

FULL MODEL ANSWER — PART (A)(III)

Suggested variable: merchandise revenue (or streaming-platform revenue) in year t. Earnings for a touring/recording act are driven substantially by merchandise sold at concerts and online, and by streaming royalties, neither of which is captured by "downloads" (a legacy/purchase-based measure) or ticket counts. Adding this variable would likely (a) improve the model's explanatory power, (b) reduce omitted-variable bias on the existing coefficients if merchandise/streaming revenue is correlated with tour activity (T_t) or album cycles (A_t), and (c) map more directly onto how a modern band actually earns money.

SOURCING NOTE

Any economically sound, well-justified variable (e.g. social media following, sponsorship income, a time trend capturing the band's career stage) earns full marks here provided the justification is clear — this is an open-ended "outside knowledge" part, not something with a single correct answer in the uploaded course material.

Mark Allocation — Part (a) (60 marks total)

Tier	% of part (a)	Criteria
Full credit	90–100%	Correctly interprets the intercept and all three slope coefficients with correct sign, magnitude and units, AND correctly assesses statistical significance using t-ratios, AND interprets $\bar{R}^2$ correctly, AND gives a well-justified expectations discussion in (ii) AND a well-justified additional variable in (iii).
Good	70–89%	Correctly interprets most coefficients and $\bar{R}^2$, but is missing significance testing or one coefficient's units/magnitude; (ii) and (iii) reasonably argued with minor gaps.
Partial	40–69%	States coefficient signs correctly but interpretation is vague (no "holding other variables constant", no units); $\bar{R}^2$ mentioned but not interpreted; (ii)/(iii) superficial.
Minimal	0–39%	Restates the equation with no real interpretation, or reverses signs/units; (ii)/(iii) missing or nonsensical.

(b) Suppose that a researcher runs a regression to examine the factors that affect take-up of private health insurance in Ireland from Quarter 1, 2003 to Quarter 4, 2016 and gets the following results (standard errors in brackets):

$$T_t = 784.3469 + 0.040072C_t + 9.218391P_t - 1.593816I_t$$

$$(0.008084) \quad (10.58059) \quad (0.707827)$$

$$\bar{R}^2 = 0.56833 \quad n = 56 \quad RSS = 76,986.76 \quad ESS = 111,648.9$$

where T_t = people with private health insurance (000s), C_t = personal consumption expenditure (€m), P_t = labour force participation rate (%), I_t = health insurance CPI (Dec 2016 = 100).

(i) Looking at this output (without carrying out any statistical tests), would you expect the regression to be significant? Explain your answer. 10%

(ii) Formally test this regression for significance at the 5% level. Explain the steps you take. 30%

FULL MODEL ANSWER — PART (B)(I)

Yes, on an informal ("eyeball") basis the regression looks likely to be jointly significant. Two things point this way even before running a formal test. First, $\bar{R}^2 = 0.568$ is moderate-to-high for a 56-quarter macro/behavioural series, implying the three regressors together explain more than half of the variation in insurance take-up. Second, informally dividing each coefficient by its standard error, the coefficient on C_t looks very large relative to its SE (roughly $0.040/0.008 \approx 5$), and I_t also looks reasonably large relative to its SE (roughly $-1.59/0.71 \approx -2.3$); only P_t looks weak relative to its SE. With at least two of the three regressors looking individually important, the joint (overall) test is very likely to come back significant.

SOURCING NOTE

This part explicitly asks for an informal assessment ("without carrying out any statistical tests") — the formal F-test with exact critical values is reserved for part (ii); credit here is for sound qualitative reasoning about $\bar{R}^2$ and rough coefficient-to-SE ratios, not for quoting exact t or F critical values.

FULL MODEL ANSWER — PART (B)(II)

Step 1 — State the hypotheses. This is a test of the overall (joint) significance of the regression:

$H_0: \beta_C = \beta_P = \beta_I = 0$ (none of the regressors jointly explain T_t)

H_1 : at least one $\beta_k \neq 0$

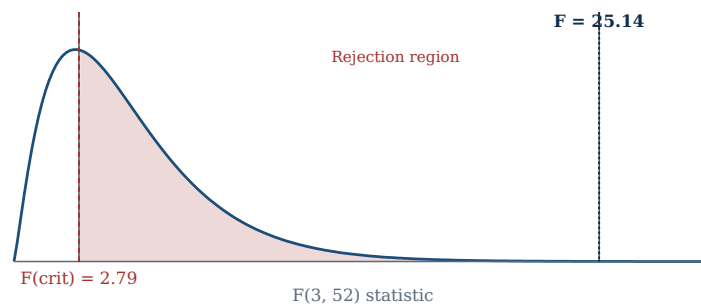
Step 2 — Choose the test statistic. The appropriate test is the F-test for overall significance:

$F = [ESS/k] / [RSS/(n-k-1)]$, which is distributed $F(k, n-k-1)$ under H_0 , where k = number of slope coefficients = 3 and $n = 56$.

Step 3 — Compute the statistic. $df_1 = k = 3$, and $df_2 = n - k - 1 = 56 - 3 - 1 = 52$.

Working — F-test for overall significance

Quantity	Value
ESS	111,648.9
RSS	76,986.76
k (df_1)	3
$n - k - 1$ (df_2)	52
ESS / k	37,216.30
RSS / ($n-k-1$)	1,480.51
F statistic	25.14
$F_{crit}(3, 52)$ at 5% (interpolated from Table A4, using $df_2 = 40 \rightarrow 2.84$ and $df_2 = 60 \rightarrow 2.76$)	≈ 2.79



The computed F statistic (25.14) lies far inside the rejection region beyond $F_{crit} \approx 2.79$.

Step 4 — Decision rule. Reject H_0 if $F > F_{crit}$. Since $25.14 > 2.79$, we reject H_0 .

Step 5 — Conclusion. There is strong evidence, at the 5% level, that the regression is jointly significant — personal consumption expenditure, labour force participation and the health insurance CPI, taken together, significantly explain variation in private health insurance take-up in Ireland over this period. This formal result confirms the informal impression from part (i).

SOURCING NOTE

The F-test formula and its use for overall significance are standard material covered directly in the EC4215 course pack. Table A4 (Critical Values of the F Distribution) is reproduced verbatim from the exam paper's own appendix; since the exact $df_2 = 52$ is not listed, the critical value is linearly interpolated between the neighbouring $df_2 = 40$ and $df_2 = 60$ columns — a standard, examiner-accepted approach when a table doesn't list the exact degrees of freedom.

Mark Allocation — Part (b) (40 marks total)

Tier	% of part (b)	Criteria
Full credit	90–100%	(i) gives sound informal reasoning using $\bar{R}^2$ and rough coefficient/SE ratios; (ii) correctly states H_0/H_1 , uses the correct F formula and correct df, computes $F = 25.14$, finds/interpolates $F_{crit} \approx 2.79$, and draws the correct reject/conclude decision.
Good	70–89%	(ii) has the correct F statistic and correct decision but a minor slip in df or critical value sourcing; (i) reasoning present but thin.
Partial	40–69%	Correct F formula but arithmetic or df errors change the numeric outcome; conclusion not clearly linked back to the hypotheses.
Minimal	0–39%	Wrong test used (e.g. a t-test in place of the F-test), or no decision/conclusion given.

Question A1 — Total: 100%

Question A2

Deciding whether to add a variable & the omitted-variable-bias result

(a) Suppose a researcher has run a regression to examine the factors affecting cinema attendances in Ireland over a year and gets the following results (standard errors in brackets):

$$CT_t = 35.3 + 11.8RF_t + 21.3AD_t + 9.5SH_t \quad (\text{Eqn. 2.1})$$

(3.8) (5.4) (3.4)

$$N = 52 \quad \text{Adjusted } R^2 = 0.582$$

Adding a variable AC_t (=1 if a major awards ceremony falls in the week):

$$CT_t = 36.8 + 12.6RF_t + 18.4AD_t + 9.3SH_t + 9.2AC_t \quad (\text{Eqn. 2.2})$$

(3.7) (4.9) (3.3) (2.1)

$$N = 52 \quad \text{Adjusted } R^2 = 0.654$$

Should the researcher include AC_t in her regression (i.e. go with Eqn. 2.2 instead of Eqn. 2.1)? Explain your answer.

70%

FULL MODEL ANSWER — PART (A)

This is a variable-inclusion decision, best answered by working systematically through the four standard specification criteria for whether a candidate variable belongs in the equation: **theory**, the **t-test**, the change in **adjusted $\bar{R}^2$** , and **bias** in the other coefficients.

1. Theory. The variable is theoretically sound: major awards ceremonies (Oscars, Golden Globes, etc.) generate substantial media coverage of films and stars, which plausibly drives short-term interest in cinema-going independent of rainfall, advertising spend, or school holidays. There is no reason to expect its place in the equation is ambiguous.

2. t-test. The t-ratio on AC_t is $9.2/2.1 = 4.38$, which comfortably exceeds any conventional critical value (e.g. 1.96 at 5%, two-tailed, for this sample size), so the coefficient is highly statistically significant and correctly signed (positive, as theory predicts).

3. Adjusted $\bar{R}^2$. Adjusted $\bar{R}^2$ rises from 0.582 to 0.654 — an increase of 0.072. Because adjusted $\bar{R}^2$ (unlike plain R^2) penalises the loss of a degree of freedom from adding a regressor, it can only rise if the new variable's t-ratio exceeds 1 in absolute value; a rise of this size, alongside a t-ratio of 4.38, is strong independent confirmation that AC_t genuinely improves the fit rather than just mechanically inflating R^2 .

4. Bias in other coefficients. Comparing Eqn. 2.1 and Eqn. 2.2, the other coefficients move noticeably when AC_t is added: RF_t rises slightly ($11.8 \rightarrow 12.6$), while AD_t falls by nearly 3 points ($21.3 \rightarrow 18.4$). A meaningful shift like this is a classic sign that AC_t was previously an **omitted variable** — award-ceremony weeks may coincide with higher advertising spend (studios advertise more around awards season), so leaving AC_t out was inflating the estimated advertising effect through omitted-variable bias.

Working — the four specification criteria applied to AC_t

Criterion	Eqn. 2.1 (without AC_t)	Eqn. 2.2 (with AC_t)	Verdict
Theory	—	Sound: media exposure → cinema-going	Pass
t-ratio on AC_t	—	9.2 / 2.1 = 4.38	Pass (highly significant)
Adjusted $\bar{R}^2$	0.582	0.654 ($\Delta = +0.072$)	Pass (improves)
Bias in other coefficients	$AD_t = 21.3$	$AD_t = 18.4$ (notable shift)	Confirms prior omission

Conclusion. All four criteria point the same way: AC_t should be included. The researcher should adopt Eqn. 2.2 over Eqn. 2.1.

SOURCING NOTE

The four-criteria framework (theory, t-test, adjusted $\bar{R}^2$, and bias in other coefficients) for deciding whether a variable belongs in an equation is taken directly from the EC4215 course pack's treatment of specification error.

Mark Allocation — Part (a) (70 marks total)

Tier	% of part (a)	Criteria
Full credit	90–100%	Works through all four criteria (theory, t-test with correct t-ratio, adjusted $\bar{R}^2$ comparison, and bias/shift in other coefficients) and reaches the correct conclusion that AC_t should be included.
Good	70–89%	Covers t-test and adjusted $\bar{R}^2$ correctly and reaches the right conclusion, but omits theory or the bias/omitted-variable link.
Partial	40–69%	Notes that adjusted $\bar{R}^2$ rose and concludes AC_t should be included, but with no t-test or reasoning beyond " R^2 went up".
Minimal	0–39%	Compares plain fit without addressing significance, or reaches the wrong conclusion.

(b) Suppose that the 'true' regression model is:

$$\text{Eqn. 2.1 } Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

where u_i is the classical disturbance term. However, suppose a researcher estimates:

$$\text{Eqn. 2.2 } Y_i = \alpha_0 + \alpha_1 X_{1i} + v_i$$

Assess the consequences of estimating the mis-specified model (Eqn. 2.2) instead of the 'true' model (Eqn. 2.1). **30%**

FULL MODEL ANSWER — PART (B)

Eqn. 2.2 omits X_2 , so its error term absorbs X_2 's effect: $v_i = \beta_2 X_{2i} + u_i$. The key issue is that X_1 and X_2 are almost always correlated in real data, which means X_1 is now correlated with the new composite error term v_i (since v_i contains X_2). This violates the classical assumption that regressors are independent of the error term.

1. Bias (and inconsistency). Because X_1 is correlated with v_i , OLS on Eqn. 2.2 produces a biased and inconsistent estimate of α_1 . The size and direction of the bias is given by the omitted-variable-bias formula:

$$E(\hat{\alpha}_1) = \beta_1 + \beta_2 \cdot (\text{Cov}(X_1, X_2) / \text{Var}(X_1))$$

so the bias term is the omitted variable's true effect (β_2) scaled by how strongly the omitted and included variables move together. The bias disappears only in two special cases: (i) $\beta_2 = 0$ (X_2 genuinely doesn't matter), or (ii) $\text{Cov}(X_1, X_2) = 0$ (X_1 and X_2 are uncorrelated) — in every other case, $\hat{\alpha}_1$ will systematically over- or under-estimate the true β_1 , and the direction depends on the signs of β_2 and the covariance.

2. Consistency does not save it. Because this correlation between X_1 and v_i does not disappear as n grows, the bias is not merely a small-sample problem: the estimator is inconsistent, so more data will not fix it.

3. Standard errors and inference are invalidated. Even where the bias happened to be small, the estimated error variance from Eqn. 2.2 is contaminated by the omitted X_2 effect, which distorts the standard error of $\hat{\alpha}_1$. This means t-tests, confidence intervals and R^2 from the mis-specified equation are all unreliable, since they are built on a variance estimate that no longer reflects the true disturbance process.

4. All other coefficients in a multi-regressor setting are also affected. More generally (beyond this simple two-variable case), omitting a relevant variable that is correlated with any included regressor biases the coefficients on all of those correlated regressors, not just the ones most obviously related to the omission — which is why the course material treats an omitted relevant variable as grounds to be suspicious of the entire equation, not just one coefficient.

Working — deriving the omitted-variable-bias result

Step	Expression
True model	$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$
Estimated (mis-specified) model	$Y_i = \alpha_0 + \alpha_1 X_{1i} + v_i$
Composite error	$\hat{v}_i = \beta_2 X_{2i} + u_i$
Bias result	$E(\alpha_1) = \beta_1 + \beta_2 \cdot \text{Cov}(X_1, X_2) / \text{Var}(X_1)$
Bias vanishes iff	$\beta_2 = 0$ OR $\text{Cov}(X_1, X_2) = 0$

SOURCING NOTE

This derivation and the resulting bias formula are taken directly from the "Consequences of an Omitted Variable" section of the EC4215 course pack, which uses this exact two-variable set-up.

Cross-reference: this is the same bias mechanism flagged qualitatively in Question A2(a), where advertising spend's coefficient shifted once the correlated, previously-omitted AC_t was added.

Mark Allocation — Part (b) (30 marks total)

Tier	% of part (b)	Criteria
Full credit	90–100%	Correctly identifies that X_1 becomes correlated with the composite error, states the bias formula correctly, identifies the two conditions under which bias vanishes, AND notes the inconsistency (not just small-sample bias) and the knock-on effect on standard errors/inference.
Good	70–89%	Correctly identifies bias and states the general direction (depends on correlation sign), but omits the inconsistency point or the standard-error/inference consequence.
Partial	40–69%	States that omitting X_2 "causes bias" without deriving why (no reference to correlation between X_1 and the error term) or without the formula.
Minimal	0–39%	Describes irrelevant-variable consequences instead of omitted-variable consequences, or gives no formal reasoning.

Question A2 — Total: 100%

Question A3

Diagnosing multicollinearity, the White test, and comparing heteroscedasticity tests

(a) A researcher examined determinants of health expenditure across Europe (HEXPTOT, GDP, POP75PROP, ALOS, NONSMOK, BEDS, etc.) and ran HEXPTOT on GDP, POP75PROP, ALOS and NONSMOK for 26 observations. EViews output (reproduced below) gave:

$R\text{-squared} = 0.986665$, $\text{Adjusted } R\text{-squared} = 0.984126$, $F\text{-statistic} = 388.4654$, $\text{Prob}(F\text{-stat}) = 0.000000$, $\text{Durbin-Watson stat} = 2.580435$.

Coefficients (t-statistics in brackets): C: 36413.90 (0.7477), GDP: 0.108243 (33.327), POP75PROP: -650.1453 (-0.3384), ALOS: 1534.125 (0.4599), NONSMOK: -600.0428 (-0.9559).

(i) Does anything in the above results suggest that this regression suffers from any econometric problems? Explain your answer. 20%

(ii) The researcher then ran the White Test and obtained a Chi-squared statistic of 25.50482, compared with a critical Chi-squared statistic (14 df, 5%) of 23.7. Does this result suggest heteroscedasticity? Explain your answer. 20%

FULL MODEL ANSWER — PART (A)(I)

The headline symptom: high R^2 but few significant t-ratios. The regression fits extremely well overall — $R\text{-squared} = 0.987$, $\text{adjusted } R\text{-squared} = 0.984$, and the overall $F\text{-statistic}$ (388.5) is highly significant ($\text{Prob} = 0.0000$). Yet of the four explanatory variables, only GDP has a statistically significant individual t-ratio (33.33); POP75PROP ($t = -0.338$), ALOS ($t = 0.460$) and NONSMOK ($t = -0.956$) are all individually insignificant even though, as a group, the four variables jointly explain almost all of the variation in health expenditure. This exact combination — a very high, highly-significant overall fit alongside mostly insignificant individual coefficients — is the textbook first symptom of **multicollinearity**: the explanatory variables are highly correlated with one another, so the regression cannot isolate each one's separate contribution even though it can explain the dependent variable very well as a group.

Why this makes economic sense here. GDP, the age-75+ population share, average length of hospital stay and the non-smoking rate are all things that move together across richer versus poorer European countries (richer countries tend to have larger elderly shares, different hospital-stay norms and different smoking rates), so a high degree of correlation among these four regressors is plausible.

Durbin-Watson. The Durbin-Watson statistic of 2.58 is close to 2 (the value indicating no first-order autocorrelation); if anything it sits slightly above 2, which would point towards very mild negative autocorrelation rather than positive autocorrelation, but this is not a serious concern here and is not the main problem flagged by this output — the multicollinearity symptom is the dominant issue.

Working — individual significance check ($t_{\text{crit}} \approx 2.08$, $\text{df} = 26 - 4 - 1 = 21$, 5% two-tail)

Variable	Coefficient	t-statistic	Individually significant?
C (intercept)	36,413.90	0.748	No
GDP	0.108243	33.327	Yes
POP75PROP	-650.1453	-0.338	No
ALOS	1,534.125	0.460	No
NONSMOK	-600.0428	-0.956	No

SOURCING NOTE

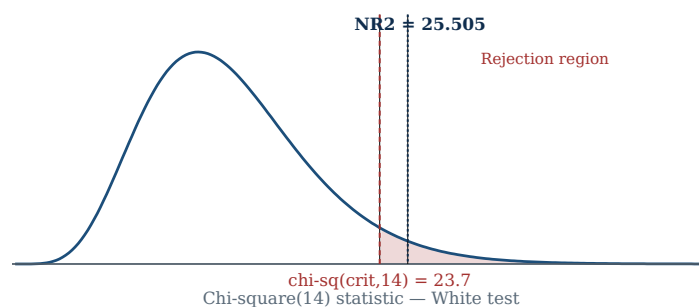
"High R^2 but few significant t-ratios" as the first diagnostic symptom of multicollinearity is listed explicitly in the EC4215 course pack's "Detection of Multicollinearity" material.

Mark Allocation — Part (a)(i) (20 marks total)

Tier	% of part	Criteria
Full credit	90–100%	Correctly identifies multicollinearity via the high- R^2 /few-significant-t-ratios pattern, cites the relevant t-statistics correctly, and gives an economically sensible reason the regressors would be correlated.
Good	70–89%	Identifies multicollinearity correctly but doesn't fully quantify which t-ratios are/aren't significant.
Partial	40–69%	Notices that "most variables are insignificant" but doesn't connect this to the high R^2 or name multicollinearity.
Minimal	0–39%	States the regression "has no problems" because R^2 is high, or misdiagnoses the issue (e.g. as autocorrelation).

FULL MODEL ANSWER — PART (A)(II)

The White test decision rule. Under the White test, the residuals from the original regression are squared and regressed on the original regressors, their squares, and their pairwise cross-products; the test statistic is NR^2 from that auxiliary regression, distributed χ^2 with degrees of freedom equal to the number of slope coefficients in the auxiliary regression. With 4 original regressors, the auxiliary equation has $4 + 4 + D(4,2) = 4 + 4 + 6 = 14$ terms, matching the 14 degrees of freedom given in the question. If the test statistic exceeds the critical χ^2 value, we reject the null hypothesis of homoscedasticity.



The White test statistic (25.505) exceeds the critical $\chi^2(14)$ value of 23.7, placing it inside the rejection region.

Working — White test decision

Quantity	Value
Test statistic (NR^2)	25.50482
Critical $\chi^2(14)$ at 5%	23.7
Decision	$25.505 > 23.7 \rightarrow \text{reject } H_0$

Conclusion. Yes — since the White test statistic (25.505) exceeds the critical value (23.7), we reject the null hypothesis of homoscedasticity at the 5% level and conclude that the regression's error variance is not constant, i.e. the residuals show evidence of heteroscedasticity. This is a separate diagnostic conclusion from the multicollinearity finding in part (i) — a regression can suffer from both problems simultaneously, and the White test specifically targets non-constant error variance rather than correlation among the regressors.

SOURCING NOTE

The White test steps and its $NR^2 \sim \chi^2$ (number of auxiliary-equation slope coefficients) decision rule are reproduced from the EC4215 course pack's "The White Test" slides.

Mark Allocation — Part (a)(ii) (20 marks total)

Tier	% of part	Criteria
Full credit	90–100%	Correctly states the decision rule (compare NR^2 to χ^2_{crit}), correctly compares 25.505 to 23.7, and correctly concludes heteroscedasticity is present.
Good	70–89%	Reaches the correct conclusion but doesn't explain why the test statistic follows a χ^2 distribution or where the 14 df come from.
Partial	40–69%	States the numbers but draws the wrong or unclear conclusion.
Minimal	0–39%	No reference to the decision rule; guesses an answer.

(b) Describe at least two different tests for detecting heteroscedasticity and outline the similarities and differences between them.

60%

FULL MODEL ANSWER — PART (B)

Test 1 — The Breusch-Pagan (BP) test. Steps: (1) estimate the original OLS regression and obtain the squared residuals; (2) regress the squared residuals on the k regressors included in the original model (or on any other variables believed to drive the error variance); (3) the null hypothesis is homoscedasticity, i.e. all slope coefficients in this auxiliary regression are jointly zero; (4) use the F-statistic from the auxiliary regression, with $(k-1)$ and $(n-k)$ degrees of freedom, to test this joint hypothesis. If the F-statistic is significant, reject homoscedasticity.

Test 2 — White's test. As set out in part (a)(ii): regress the squared residuals on the original regressors, their squares, AND their pairwise cross-products; take NR^2 from that auxiliary regression as the test statistic, which is χ^2 -distributed with degrees of freedom equal to the number of auxiliary-equation slope coefficients. Reject homoscedasticity if NR^2 exceeds the critical χ^2 value.

A third option worth naming — the Goldfeld-Quandt (GQ) test. Steps: (1) order the observations by the size of the variable thought to drive the heteroscedasticity; (2) drop the middle d observations; (3) run two separate regressions — one on the low- X subsample, one on the high- X subsample; (4) obtain RSS_1 (low- X) and RSS_2 (high- X); (5) form the ratio $GQ = [RSS_2/df] / [RSS_1/df]$, an F-statistic with $(0.5)(N-d-2k)$ degrees of freedom in numerator and denominator; (6) if GQ exceeds F_{crit} , reject homoscedasticity.

Working — comparing the three tests

Feature	Breusch-Pagan	White	Goldfeld-Quandt
Auxiliary regressors used	Original k regressors (or a specified set)	Original regressors + their squares + pairwise cross-products	None — splits the sample instead of running an auxiliary regression
Test statistic	F-statistic, $(k-1, n-k)$ df	NR^2 , χ^2 with df = number of auxiliary slope coefficients	F-statistic, ratio of two subsample RSS
Needs a specified functional form for the heteroscedasticity?	Yes — assumes error variance is (roughly) linear in the chosen regressors	No — more general/flexible; captures nonlinear and interaction patterns via squares and cross-products	Yes — needs the analyst to pick and order by the variable believed to drive the variance, and to choose d
Sample requirement	Uses the whole sample	Uses the whole sample	Requires ordering the data and dropping the middle d observations, so needs a reasonably large n

Similarities. All three share the same underlying logic: heteroscedasticity means the error variance is systematically related to one or more variables, so each test looks for a statistically significant relationship between some measure of the error's size (squared residuals, or RSS across subsamples) and the suspected driver(s) of that variance. All three end in the same type of decision rule: compare a

computed test statistic to a critical value and reject homoscedasticity if the statistic is large enough.

Key difference. White's test is the most general of the three because it doesn't require the analyst to specify in advance which variable, or what functional form, drives the heteroscedasticity — it lets the data reveal any relationship (including nonlinear and interaction effects) between the regressors and the error variance. The Breusch-Pagan test is more restrictive because it assumes a specific (typically linear) relationship between the error variance and the chosen regressors. The Goldfeld-Quandt test takes a completely different approach — comparing residual variance across two ordered subsamples rather than running an auxiliary regression — but requires the analyst to correctly guess, in advance, which variable the variance depends on and to choose how many middle observations (d) to drop, both of which are somewhat arbitrary choices that can affect the result.

SOURCING NOTE

All three tests, their exact steps, and the direct statement that "the White test is more general and more flexible than the BP test" are taken verbatim in substance from the EC4215 course pack's heteroscedasticity-detection material.

Mark Allocation — Part (b) (60 marks total)

Tier	% of part (b)	Criteria
Full credit	90–100%	Correctly describes at least two tests step-by-step (including the correct auxiliary regression/test statistic/distribution for each), AND gives at least one genuine similarity AND one genuine difference between them (e.g. White's generality vs. BP's need for a specified functional form).
Good	70–89%	Describes two tests reasonably accurately but the similarities/differences discussion is thin or one-sided.
Partial	40–69%	Names two tests and gives a rough description of one, but gets a step wrong (e.g. wrong test statistic or distribution) or omits comparison entirely.
Minimal	0–39%	Names tests without describing their steps, or describes a different econometric problem (e.g. autocorrelation tests) instead.

Question A3 — Total: 100%